# SHANICE T. BAILEY

Environmental Learning Designer & Trainer
Climate & Ocean Science Education
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## Professional Profile

Environmental learning designer and climate scientist with 6+ years of experience designing curricula, workshops, and training programs that translate complex climate and ocean science into accessible learning experiences. Proven track record developing multi-week curricula, facilitating virtual and in-person workshops, and integrating real scientific data into education programs. Seeking remote or hybrid roles in curriculum development, climate education, and environmental training.

## TEACHING/OUTREACH EXPERIENCE

### PyClub Afterschool Program | Virtual

**Curriculum Co-Designer & Instructor** | 2020 – 2021

- Co-designed and implemented an **8-week afterschool curriculum** teaching Python through climate and ocean science for high school students.
- Led design of a full weekly lesson integrating real environmental datasets, data visualization, and scientific interpretation.
- Applied scaffolded, inquiry-based pedagogy to support student engagement and retention.
- Conducted **multiple formative surveys** to assess student experience; feedback highlighted increased confidence using Python and strong engagement with climate applications.
- Curriculum materials documented and made publicly available for replication.

♣ Curriculum library: https://pyclub-cu.github.io/library.html

### Coastal Ocean Environment Summer School (COESSING)

**Curriculum Contributor & Instructor** | Summer 2023

- Co-developed curriculum for an international summer school advancing ocean science education in West Africa.
- Helped design program schedule for virtual components and delivered lectures on research computing tools (JupyterLab, Xarray).
- Adapted advanced scientific concepts for a diverse, interdisciplinary audience.

### Software Carpentry Program

**Certified Instructor & Curriculum Developer** | 2021 – Present

- Co-developed and co-instructed remote Python courses for graduate students across scientific disciplines.

- Designed hands-on lessons, practice exercises, and live-coding demonstrations using real scientific datasets.
- Delivered instruction remotely using Zoom, Zulip, and Google Colab.

### Girls Who Code
**Instructor** | 2019

- Designed full course syllabus and instructional materials introducing coding concepts to students.
- Delivered lectures and developed practice problems to reinforce computational thinking.

### Curriculum Innovation Grant Fellow
**Curriculum Developer** | 2020

- Designed a technical training curriculum focused on expanding research computing skills.
- Collaborated with institutional partners to pilot the curriculum through facilitated workshops.

## WORKSHOPS, TRAINING & PUBLIC ENGAGEMENT

- **GFDL Graduate School Application Workshop** (2025): Co-facilitated professional development workshop for summer interns; developed presentation materials and provided individualized feedback.
- **LEAP Train-the-Trainer Bootcamp** (2023): Provided instructional and technical support for a two-day climate data science training workshop.
- **NSBP–HGS Mentoring Program** (2020–2023): Mentored middle and high school students from underrepresented communities in STEM.
- **Public Panels & Science Outreach**: Participated in public discussions on climate science and representation in STEM.

## RESEARCH & TECHNICAL EXPERIENCE

Princeton University | Geophysical Fluid Dynamics Laboratory
**Postdoctoral Research Associate**, Atmospheric and Oceanic Sciences **2024 – Present**

Columbia University Graduate of School of Arts & Science
**Graduate Research Assistant (PhD)**, Earth and Environmental Sciences **2018 – 2024**

- Conducted research on ocean circulation, marine heatwaves, and climate variability using numerical models and observational datasets.
- Extensive experience working with large climate datasets and translating results for diverse audiences.
  *(Publications and full research details available upon request or on website.)*

## EDUCATION

Columbia University Graduate of School of Arts & Science
**PhD**, Earth and Environmental Sciences **2024**

Dissertation: *Water mass transformation through the lens of numerical models and observations*
Advisor: Dr. Ryan P. Abernathey

**MPhil**, Earth and Environmental Sciences **2021**
**MA**, Earth and Environmental Sciences **2020**

Columbia University School of Professional Studies
**Postbaccalaureate Student** **2017 – 2018**

Boston University
**B.A., Environmental Sciences & Wildlife- Ecology/Management** **2012 – 2016**
Thesis: "Feeding Behavior of African Elephants (*Loxodonta africana)* in Tarangire-Manyara ecosystem"

## PUBLICATIONS

**Bailey, S. T.**, Jones, C. S., Abernathey, R. P.: North Atlantic Subtropical Mode Water transformation by near surface eddy mixing in high resolution climate models, *In preparation for the Journal of Geophysical Research*.

**Bailey, S. T.** , Drake, H. F., Gruenburg, L. K., Abernathey, R. P., & Torres, M. I. (2025). The thermodynamics of the 2023 Gulf of Mexico Marine heatwave. Geophysical Research Letters, 52, e2024GL111768, https://doi.org/10.1029/2024GL111768, 2025 .

**Bailey, S. T.**, Jones, C. S., Abernathey, R. P., Gordon, A. L., and Yuan, X.: Water mass transformation variability in the Weddell Sea in ocean reanalyses, Ocean Sci., 19, 381–402, https://doi.org/10.5194/os-19-381-2023, 2023.

*Professional Feature* – ArcGIS StoryMap featured on *Estimating the Circulation and Climate of the Ocean* group's Featured Publications page. Highlights work from **Bailey, S.T., et al., 2023**.

## CORE SKILLS

Curriculum design · Learning experience design · Climate & ocean science communication · Workshop facilitation · Program evaluation · Python-based data analysis · Virtual instruction · Instructional technology (Zoom, LMS, Jupyter, Google Colab)